PlusMinus Analytics:

What "I" Did

- I developed a basketball analytics prototype that connects shooting results to a suggested practice plan.
- I used Python to clean historical shot data and aggregate it for comparisons.
- I also produced the product requirements, feature blueprint, user flows, and roadmap.

What the documents do

- The Product Requirements Document (PRD) defines the problem, intended user, requirements, and acceptance criteria.
- The feature blueprint explains each capability, while the user flows show how someone moves from reviewing results to choosing a next action.
- The six-month roadmap proposes the order of future development. It is a planning horizon, not six months of completed work.

What makes it personalized?

- The plan uses the selected player's results to find the lowest-accuracy zone with at least 20 attempts.
- It assigns approximately 20% of the chosen time to warm-up, 60% to focused practice, and the remainder to review.
- The recommendation follows explicit rules. It does not use live AI or diagnose shooting technique.

How did I get the numbers?

- I counted seven entities: players, sessions, shot maps, events, coaches, ratings, and plans.
- The player, coach, and training dashboards each contain five visuals, giving 15 visuals across three dashboards.
- Five simulated API contracts cover players, sessions, shot maps, ratings, and plans. They run in the browser rather than on a live backend.
- The browser reads JSON collections, and a separate SQLite artifact shows the corresponding database structure.

Case study + impact

- The case study includes architecture diagrams, competitive analysis, proposed KPIs, and a go-to-market strategy.
- It explains the implementation and next steps. I have not measured a threefold increase in recruiter engagement.

Interview examples:

Data quality: fixing the era field

- The era classification could incorrectly use team names or misinterpret season identifiers.
- The correction prioritizes explicit season information and uses game dates only as a fallback, including a 2020 bubble exception.
- Rebuilt exports and regression tests support the corrected grouping. My takeaway is that a plausible chart can still rely on incorrect data.

Data discovery: know the source

- The historical sample contains 79,763 shots across 10 selected NBA players; it does not represent the entire league.
- The training prototype uses three fictional players, 24 sessions, and 1,440 synthetic shots.
- I kept the sources separate and documented what each record represents and which conclusions the data can support.

Validation: can I trust the metric?

- I calculate overall accuracy as total makes divided by total attempts, rather than averaging session percentages.
- For example, sessions of 1/2 and 9/10 combine to 10/12, or 83.3%. With no attempts, accuracy stays unknown.
- The tests check record relationships, metric totals, plan durations, and API responses. All four Node tests passed during the project review.

Product thinking: what comes next?

- I narrowed the proposed next release to logging a session, reviewing results, choosing a plan, and recording a follow-up.
- The requirements include reliable saving, duplicate prevention, and controls that keep each user's records private.
- I would measure whether users complete the workflow unaided, understand the plan, and return. The pilot and production backend remain proposed work.

Why this fits Vanguard

- This project connects data discovery, clear business rules, validation, and communication of limitations.
- I used AI coding assistance and should distinguish what I personally built, reviewed, and validated when discussing these examples.